

1 Identification of the substance/mixture and of the company/undertaking**· Product identifier****· Trade name:****Herculan BA 2000 { B }****· Article number:**

HE_801302/14

· Relevant identified uses of the substance or mixture and uses advised against

No further relevant information available.

· Application of the substance / the mixture

Component or ready-mix for in-situ creation of flooring / roofing / adhesives / resins - materials.
The final product is created only upon curing - Chemical properties of the starting materials in section 3.

· Details of the supplier of the safety data sheet**· Manufacturer/Supplier:****HERCULAN B.V.**

c/o EPUTAN Kunststofftechnik B.V.

Energieweg 6

4231 DJ Meerkerk

The Netherlands

Phone +31 183 354700

Fax: +31 183 354740

e-mail: info@herculan.com

· Further information obtainable from:

Environmental Department

· Emergency telephone number:

+49 (0) 6131 19240 [24 h - 365 d] - Giftinformationszentrale Mainz
or
+31 (0) 183 354 700 [Mo - Fr. 8 - 17 o'clock] - HERCULAN

2 Hazards identification**· Classification of the substance or mixture**

Acute Tox. 4 H302 Harmful if swallowed.
Acute Tox. 5 H333 May be harmful if inhaled.
Skin Corr. 1B H314 Causes severe skin burns and eye damage.
Eye Dam. 1 H318 Causes serious eye damage.
Skin Sens. 1 H317 May cause an allergic skin reaction.
Aquatic Acute 3 H402 Harmful to aquatic life.
Aquatic Chronic 3 H412 Harmful to aquatic life with long lasting effects.

· Label elements**· GHS label elements****· Hazard pictograms**

The product is classified and labelled according to the Globally Harmonised System (GHS).



GHS05 GHS07

· Signal word

Danger

· Hazard-determining components of labelling:

benzyl alcohol
1,3-Benzenedimethanamine
3-aminomethyl-3,5,5-trimethylcyclohexylamine
3,6,9-triazaundecamethylenediamine
H302 Harmful if swallowed.
H333 May be harmful if inhaled.
H314 Causes severe skin burns and eye damage.
H317 May cause an allergic skin reaction.
H412 Harmful to aquatic life with long lasting effects.
P260 Do not breathe dusts or mists.

· Hazard statements**· Precautionary statements**

(Contd. on page 2)

Trade name: **Herculan BA 2000 { B }**

(Contd. of page 1)

P303+P361+P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower.

P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P362+P364 Take off contaminated clothing and wash it before reuse.

P405 Store locked up.

P501 Dispose of contents/container in accordance with local/regional/national/international regulations.

· **Other hazards**· **Results of PBT and vPvB assessment**· **PBT:**

This product contains no substances assessed as PBT at concentrations of 0.1% or higher.

· **vPvB:**

This product contains no substances rated as vPvB at concentrations of 0.1% or higher.

* **3 Composition/information on ingredients**· **Chemical characterisation: Mixtures**· **Description:**

There are no additional ingredients present which are classified as hazardous to health or the environment and on this basis need to be mentioned in this section.

· **Dangerous components:**

100-51-6	benzyl alcohol	25-100%
	⚠ Acute Tox. 4, H302; Eye Irrit. 2, H319; Skin Sens. 1B, H317	
1477-55-0	1,3-Benzenedimethanamine	10-<25%
	⚠ Skin Corr. 1B, H314; Eye Dam. 1, H318; ⚠ Acute Tox. 4, H332; Skin Sens. 1, H317; Aquatic Acute 3, H402; Aquatic Chronic 3, H412	
2855-13-2	3-aminomethyl-3,5,5-trimethylcyclohexylamine	10-<25%
	⚠ Skin Corr. 1B, H314; Eye Dam. 1, H318; ⚠ Acute Tox. 4, H302; Skin Sens. 1A, H317	
	EMMA - Amino KAT	5-<10%
	⚠ Acute Tox. 4, H302; Skin Irrit. 2, H315; Eye Irrit. 2, H319	
112-57-2	3,6,9-triazaundecamethylenediamine	5-<10%
	⚠ Skin Corr. 1B, H314; ⚠ Aquatic Chronic 2, H411; ⚠ Acute Tox. 4, H302; Acute Tox. 4, H312; Skin Sens. 1, H317	
1226892-44-9	Fatty acids, C18 unsaturated, reaction products with triethylenetetramine	≥0,1-<0,25%
	⚠ Skin Corr. 1C, H314; Eye Dam. 1, H318; ⚠ Aquatic Acute 1, H400; Aquatic Chronic 1, H410; ⚠ Skin Sens. 1, H317	
	polymeres Polyamidoamin	≥0,1-<0,25%
	⚠ Skin Corr. 1C, H314; Eye Dam. 1, H318; ⚠ Aquatic Acute 1, H400; Aquatic Chronic 1, H410; ⚠ Skin Sens. 1, H317	

· **SVHC**· **Additional information:**

Acute toxicity estimates

	Oral	Inhalative	Dermal
CAS 100-51-6	1230 mg/kg	4,188 mg/l	mg/kg

For the wording of the listed hazard phrases refer to section 16.

* **4 First aid measures**· **Description of first aid measures**· **General information:**

Immediately remove any clothing soiled by the product.



Symptoms of poisoning may even occur after several hours; therefore medical observation for at least 48 hours after the accident.

Take affected persons out into the fresh air.

Do not leave affected persons unattended.

Seek medical treatment.

Medical supervision for at least 48 hours.

Show this safety data sheet to the doctor on duty.

Protection of first aiders: No action should be taken if there is a risk of personal accidents or in case of insufficient training. This can be dangerous for the person applying mouth-to-mouth resuscitation.

(Contd. on page 3)

Trade name: Herculan BA 2000 { B }

(Contd. of page 2)

If the possibility of exposure exists, consult section 8 for specific personal protective equipment.

First-aiders should remember to protect themselves and wear the recommended protective clothing.

No action should be taken if there is a risk of personal accidents or in case of insufficient training.

· After inhalation:

Supply fresh air and to be sure call for a doctor.

In case of unconsciousness place patient stably in side position for transportation.

Seek medical treatment.

Use a respiratory bag or breathing device.

· After skin contact:

Immediately wash with water and soap and rinse thoroughly.

Immediately rinse with water.

Seek medical treatment.

· After eye contact:

Rinse opened eye for several minutes under running water. Then consult a doctor.

Protect unharmed eye.

As far as easy to do, remove any contact lenses.

· After swallowing:

Do not induce vomiting; call for medical help immediately.

Call for a doctor immediately.

Drink plenty of water and provide fresh air. Call for a doctor immediately.

A person vomiting while laying on their back should be turned onto their side.

Clean carefully or rinse mouth with water.

Keep breathing passages clear.

Never allow an unconscious person to drink (or eat).

Take patient to a hospital immediately.

· Information for doctor:**· Most important symptoms and effects, both acute and delayed**

No further relevant information available.

· Indication of any immediate medical attention and special treatment needed

No further relevant information available.

5 Firefighting measures**· Extinguishing media****· Suitable extinguishing agents:**

CO2, powder or water spray. Fight larger fires with water spray or alcohol resistant foam.

Use fire extinguishing methods suitable to surrounding conditions.

· Special hazards arising from the substance or mixture

During heating or in case of fire poisonous gases are produced.

Formation of toxic gases is possible during heating or in case of fire.

Carbon monoxide (CO)

· Advice for firefighters**· Protective equipment:**

Mount respiratory protective device.

Wear self-contained respiratory protective device.

Do not inhale explosion gases or combustion gases.

Wear fully protective suit.

PSA 52 / PSA 55 / PSA 56 / PSA 57

· Additional information

Dispose of fire debris and contaminated fire fighting water in accordance with official regulations.

Collect contaminated fire fighting water separately. It must not enter the sewage system.

6 Accidental release measures**· Personal precautions, protective equipment and emergency procedures**

Mount respiratory protective device.

Remove persons from danger area.

Wear protective equipment. Keep unprotected persons away.

(Contd. on page 4)

Trade name: Herculan BA 2000 { B }

(Contd. of page 3)

· Environmental precautions:*Keep people at a distance and stay on the windward side.**Wear protective clothing.**Do not allow product to reach sewage system or any water course.**Inform respective authorities in case of seepage into water course or sewage system.**In case of seepage into the ground inform responsible authorities.**Do not allow to enter sewers/ surface or ground water.**Prevent further leaks and spills if safe to do so. In case of significant leaks that cannot be contained, inform the local authorities.**If the product contaminates rivers, lakes or sewers, notify the respective authorities.***· Methods and material for containment and cleaning up:***Let reaction take place at least 30 minuten.**Use neutralising agent.**Dispose contaminated material as waste according to section 13.**Ensure adequate ventilation.**Scoop up into lidded drums and render harmless.**Store in suitable and closed containers for disposal.***· Reference to other sections***See Section 7 for information on safe handling.**See Section 8 for information on personal protection equipment.**See Section 13 for disposal information.**** 7 Handling and storage****· Handling:***Use according to accepted rules and practices related to industrial hygiene and safety. After working with this product, wash face, hands and exposed skin thoroughly. Remove contaminated clothing and protective equipment before going to the canteen. Do not eat, drink or smoke during use. Do not allow contaminated work clothing to leave the workplace. Wash hands before each work break and immediately after using the product. Wash hands before each work break and at the end of the working day.***· Precautions for safe handling***Provide eyewash facilities and safety showers close to the workplace.**Thorough dedusting.**Ensure good ventilation/exhaustion at the workplace.***· Information about fire - and explosion protection:***Keep respiratory protective device available.***· Conditions for safe storage, including any incompatibilities****· Storage:****· Requirements to be met by storerooms and receptacles:***No special requirements.***· Information about storage in one common storage facility:***Store away from oxidising agents.**Store away from foodstuffs.***· Further information about storage conditions:***Protect from frost.**Keep container tightly sealed.**Store in dry conditions.**Store in a cool place.***· Specific end use(s)***No further relevant information available.**** 8 Exposure controls/personal protection****· Additional information about design of technical facilities:***Keep respiratory protective device available.***· Control parameters****· Ingredients with limit values that require monitoring at the workplace:***The product does not contain any relevant quantities of materials with critical values that have to be monitored at the workplace.*

(Contd. on page 5)

Trade name: Herculan BA 2000 { B }

(Contd. of page 4)

· DNELs**100-51-6 benzyl alcohol**

Oral	DNEL ACUTE / SHORT	20 mg/kg lg/d (Consumer system-related)
	DNEL CHRONIC / LONG	4 mg/kg lg/d (Consumer system-related)
Dermal	DNEL ACUTE / SHORT	20 mg/kg lg/d (Consumer system-related)
		40 mg/kg lg/d (Worker system-related)
Inhalative	DNEL CHRONIC / LONG	4 mg/kg lg/d (Consumer system-related)
		8 mg/kg lg/d (Worker system-related)
	Shortterm value	27 mg/m ³ (Consumer system-related)
		110 mg/m ³ (Worker system-related)
	Longterm value	5,4 mg/m ³ (Consumer system-related)
		22 mg/m ³ (Worker system-related)

1477-55-0 1,3-Benzenedimethanamine

Dermal	DNEL CHRONIC / LONG	0,33 mg/kg lg/d (Worker system-related)
Inhalative	Shortterm value	0,2 mg/m ³ (Worker local)
	Longterm value	1,2 mg/m ³ (Worker system-related)

2855-13-2 3-aminomethyl-3,5,5-trimethylcyclohexylamine

Oral	Longterm value	0,3 mg/kg (Consumer system-related)
	DNEL ACUTE / SHORT	0,3 mg/kg lg/d (Consumer system-related)
	DNEL CHRONIC / LONG	0,3 mg/kg lg/d (Consumer system-related)
Inhalative	Shortterm value	0,073 mg/m ³ (Worker local)
		0,073 mg/m ³ (Worker system-related)
	Longterm value	0,073 mg/m ³ (Worker local)

EMMA - Amino KAT

Dermal	DNEL ACUTE / SHORT	0,6 mg/kg lg/d (Worker system-related)
	DNEL CHRONIC / LONG	0,15 mg/kg lg/d (Worker system-related)
Inhalative	Shortterm value	0,52 mg/m ³ (Worker system-related)
	Longterm value	0,13 mg/m ³ (Worker system-related)

· PNECs**100-51-6 benzyl alcohol**

Soil	5,27 mg/kg (PNEC Freshwater Sediment)
	0,456 mg/kg (PNEC Soil)
	0,527 mg/kg (PNEC Seawater Sediment)
Water	2,3 mg/l (PNEC Intermittent freshwater release)
	0,1 mg/l (PNEC Seawater)
	39 mg/l (PNEC Wastewater treatment)
	1 mg/l (PNEC Freshwater)

1477-55-0 1,3-Benzenedimethanamine

Soil	12,4 mg/kg (PNEC Freshwater Sediment)
	2,44 mg/kg (PNEC Soil)
	1,24 mg/kg (PNEC Seawater Sediment)
Water	0,152 mg/l (PNEC Intermittent freshwater release)
	0,009 mg/l (PNEC Seawater)
	10 mg/l (PNEC Wastewater treatment)
	0,094 mg/l (PNEC Freshwater)

2855-13-2 3-aminomethyl-3,5,5-trimethylcyclohexylamine

Soil	5,784 mg/kg (PNEC Freshwater Sediment)
	1,121 mg/kg (PNEC Soil)

(Contd. on page 6)

Trade name: Herculan BA 2000 { B }

(Contd. of page 5)

Water	0,578 mg/kg (PNEC Seawater Sediment)
	0,006 mg/l (PNEC Seawater)
	3,18 mg/l (PNEC Wastewater treatment)
	0,06 mg/l (PNEC Freshwater)
EMMA - Amino KAT	
Water	0,008 mg/l (PNEC Seawater)
	0,2 mg/l (PNEC Wastewater treatment)
	0,084 mg/l (PNEC Freshwater)

Additional information:

The lists valid during the making were used as basis.

Exposure controls**Personal protective equipment:****General protective and hygienic measures:**

Keep away from foodstuffs, beverages and feed.
 Immediately remove all soiled and contaminated clothing
 Wash hands before breaks and at the end of work.
 Store protective clothing separately.
 Avoid contact with the eyes.
 Avoid contact with the eyes and skin.
 Do not eat, drink, smoke or sniff while working.

Respiratory protection:

Only during spraying without adequate removal by suction.

Use suitable respiratory protective device in case of insufficient ventilation.

Use suitable respiratory protective device when high concentrations are present.

Recommended filter device for short term use:

Air concentrations would be kept below exposure guidelines. When airborne concentrations may exceed exposure guidelines, use an approved air-purifying respirator fitted with a filter for organic vapours and particles. In circumstances where the concentration in the air may exceed the level for which an air-purifying respirator is effective, use a compressed air respirator (type: positive pressure) (air hose-fed or self-contained respirator). In emergencies or in conditions where the concentration in air is not known, use an approved compressed air breathing apparatus (type: positive pressure) or air hose-fed respirator (type: positive pressure). Use EC-approved respirator: Filter for organic vapours with a particulate pre-filter for highly toxic substances, type AP3 (must comply with Standard EN 14387).

Protection of hands:

Protective gloves

The glove material has to be impermeable and resistant to the product/ the substance/ the preparation.

Material of gloves

The selection of the suitable gloves does not only depend on the material, but also on further marks of quality and varies from manufacturer to manufacturer. As the product is a preparation of several substances, the resistance of the glove material can not be calculated in advance and has therefore to be checked prior to the application.

Penetration time of glove material

The exact break through time has to be found out by the manufacturer of the protective gloves and has to be observed.

The determined penetration times according to EN 16523-1:2015 are not performed under practical conditions. Therefore a maximum wearing time, which corresponds to 50% of the penetration time, is recommended.

As protection from splashes gloves made of the following materials are suitable:

Nitrile rubber, NBR

Natural rubber, NR

Not suitable are gloves made of the following materials:

Leather gloves

Strong material gloves

Eye protection:

Tightly sealed goggles

(Contd. on page 7)

Trade name: **Herculan BA 2000 { B }**

(Contd. of page 6)

· **Body protection:**

(Protective) work clothing

- Protective work clothing (long trousers, long-sleeved shirt). Avoid exposed skin, even in hot weather.

- Depending on processing: use spray-tight trousers or spray-tight overalls Overall (preferably thick cotton) or Tyvek-Pro Tech 'C', TyvekPro Tech 'F' fully protective single-use clothing.

· **Limitation and supervision of exposure into the environment**

Storage of substances in tightly closed packaging

Emissions from ventilation or processing equipment should be monitored to ensure they meet the requirements of environmental protection legislation. In some cases, gas scrubbers, filters or technical modifications to process equipment are needed to reduce emissions to acceptable levels. Avoid spreading spilled material and waste material and prevent it from coming into contact with soil, waterways, drains and sewers.

* **9 Physical and chemical properties**· **Information on basic physical and chemical properties**· **General Information**· **Appearance:**

Form:	Fluid
Colour:	Yellowish
Odour:	Characteristic
Odour threshold:	Not determined

· **pH-value:** Mixture is non-polar/aprotic.

· **Change in condition**

Melting point/freezing point:	No test data available
Initial boiling point and boiling range:	156 °C

· **Flash point:** >100 °C

· **Flammability** Not determined

· **Auto-ignition temperature:** 382 °C

· **Decomposition temperature:** No test data available

· **Ignition temperature:** Product is not selfigniting.

· **Explosive properties:** Product does not present an explosion hazard.
No test data available

· **Explosion limits:**

Lower:	1,3 Vol %
Upper:	13 Vol %

· **Vapour pressure at 20 °C:** 0,1 hPa

· **Vapour pressure at 50 °C:** 0,7 hPa

· **Density at 20 °C:** 1,009 g/cm³

· **Relative density** >> density

· **Vapour density** Not applicable

· **Evaporation rate** Not applicable.

· **Solubility in / Miscibility with water:**

Soluble.

· **Partition coefficient: n-octanol/water:** No test data available

(Contd. on page 8)

Trade name: **Herculan BA 2000 { B }**

(Contd. of page 7)

· **Viscosity:****Dynamic at 20 °C:**

100 mPas

Kinematic:

Not applicable

· **Solvent content:****Solids content:**

100,0 %

· **Other information**

No further relevant information available.

* **10 Stability and reactivity**· **Reactivity**

No further relevant information available.

· **Chemical stability**

Stable under recommended storage conditions. See Section 7, Storage.

· **Thermal decomposition / conditions to be avoided:**

Extreme temperatures and direct sunlight.

Exposure to air or moisture for prolonged periods.

· **Possibility of hazardous reactions**

No dangerous reactions known.

· **Conditions to avoid**

No further relevant information available.

· **Incompatible materials:**

No further relevant information available.

· **Hazardous decomposition products:**

No dangerous decomposition products known.

* **11 Toxicological information**· **Information on toxicological effects**· **Acute toxicity**

ATE values > 2000 mg/kg will have no impact on classification.

· **LD/LC50 values relevant for classification:****ATE (Acute Toxicity Estimates)**

Oral	Acute orale toxiciteit	615 mg/kg
Dermal	Acute dermale toxiciteit	16.094 mg/kg
Inhalative	Acute inhalatief toxiciteit	8,78 mg/l

100-51-6 benzyl alcohol

Oral	Acute orale toxiciteit	500 mg/kg (ATE)
	OECD 423 (LD50)	1.230 mg/kg (rat)
Dermal	OECD 402 (LD50)	2.000 mg/kg (rabbit)
Inhalative	OECD 403 (LC50)	4.178 mg/l (rat) (04 h)

1477-55-0 1,3-Benzenedimethanamine

Oral	OECD 423 (LD50)	940 mg/kg (rat)
Dermal	OECD 402 (LD50)	2.000 mg/kg (rabbit)
Inhalative	Acute inhalatief toxiciteit	1,5 mg/l (ATE)
	OECD 403 (LC50)	2,4 mg/l (rat) (04 h)

2855-13-2 3-aminomethyl-3,5,5-trimethylcyclohexylamine

Oral	Acute orale toxiciteit	500 mg/kg (ATE)
	OECD 423 (LD50)	1.030 mg/kg (rat)
Dermal	OECD 402 (LD50)	>2.000 mg/kg (rat)
Inhalative	OECD 403 (LC50)	5,01 mg/l (rat)

EMMA - Amino KAT

Oral	Acute orale toxiciteit	500 mg/kg (ATE)
	OECD 401 (LD50)	2.169 mg / kg (rat)
Irritation of skin	OECD 404	(rabbit)

(Contd. on page 9)

Trade name: **Herculan BA 2000 { B }**

(Contd. of page 8)

	OECD 406	(Guinea pig)
112-57-2 3,6,9-triazaundecamethylenediamine		
Oral	Acute orale toxiciteit	500 mg/kg (ATE)
Dermal	Acute dermale toxiciteit	1.100 mg/kg (ATE)

· **Primary irritant effect:**· **Skin corrosion/irritation**

Caustic effect on skin and mucous membranes.

· **Serious eye damage/irritation**

Strong caustic effect.

Strong irritant with the danger of severe eye injury.

· **Respiratory or skin sensitisation**

Sensitisation possible through skin contact.

· **Additional toxicological information:**

The product shows the following dangers according to the calculation method of the General EU Classification Guidelines for Preparations as issued in the latest version:

Harmful

Corrosive

Irritant

Swallowing will lead to a strong caustic effect on mouth and throat and to the danger of perforation of esophagus and stomach.

* **12 Ecological information**· **Toxicity**· **Aquatic toxicity:****100-51-6 benzyl alcohol**

OECD 202 (EC50/EL50)	35 mg/l (Anabaena variabilis)
	23 mg/l (Daphnia Magna) (48 h)
OECD 203 (LC50/LL50)	10 mg/l (Lepomis macrochirus (Zonnebaars)) (96 h)
	460 mg/l (Pimephales promelas) (96 h)
OECD 209 (EC50/EL50)	2.100 mg/l

1477-55-0 1,3-Benzenedimethanamine

NOEC	4,7 mg/l (Daphnia Magna) (21 d)
	10,5 mg/l (Pseudokirchneriella subcapitata (algen)) (72 h)
	33 mg/l (scenedesmus subspicatus)
OECD 201 (ErC50/ErL50)	33,3 mg/l (scenedesmus subspicatus)
OECD 202 (EC0)	8,9 mg/l (Daphnia Magna) (48 h)
OECD 202 (EC100)	28 mg/l (Daphnia Magna) (48 h)
OECD 202 (EC50/EL50)	15,2 mg/l (Daphnia Magna) (48 h)
	12 mg/l (scenedesmus subspicatus) (72 h)
OECD 203 (LC50/LL50)	87,6 mg/l (Oryzias latipes (Ricefish)) (96 h)

2855-13-2 3-aminomethyl-3,5,5-trimethylcyclohexylamine

EbC50	37 mg/l (scenedesmus subspicatus) (72 h)
EC10	1.120 mg/l (Bacteria) (18 h)
OECD 202 (EC50/EL50)	23 mg/l (Daphnia Magna) (48 h)
	37 mg/l (scenedesmus subspicatus) (72 h)
OECD 203 (LC50/LL50)	110 mg/l (Brachydanio rerio (zebravis).) (96 h)
OECD 303 A	42 % (3 h)

EMMA - Amino KAT

OECD 201 (ErC50/ErL50)	84 mg/l (Desmodesmus subspicatus) (72 h)
OECD 203 (LC50/LL50)	175 mg/l (Cyprinus carpio (Karper)) (96 h)
OECD 471	(Ames Test (Salmonella/microsome test))

(Contd. on page 10)

Trade name: Herculan BA 2000 { B }

(Contd. of page 9)

· Persistence and degradability**1477-55-0 1,3-Benzenedimethanamine**

OECD 301 B | 49 % (28 d)

2855-13-2 3-aminomethyl-3,5,5-trimethylcyclohexylamine

OECD 301 A | 8 % (28 d)

EMMA - Amino KAT

OECD 301 D | % (Activated sludge) (28 d)

· Behaviour in environmental systems:**· Bioaccumulative potential****100-51-6 benzyl alcohol**

OECD 117 (log Pow) - Bioaccumulation | 1,1 (n-octanol/water)

1477-55-0 1,3-Benzenedimethanamine

OECD 117 (log Pow) - Bioaccumulation | 0,18

2855-13-2 3-aminomethyl-3,5,5-trimethylcyclohexylamine

OECD 117 (log Pow) - Bioaccumulation | 0,79 (n-octanol/water)

· Mobility in soil

No further relevant information available.

· Ecotoxicological effects:**· Remark:**

Harmful to fish

· Additional ecological information:**· General notes:**

Water hazard class 2 (German Regulation) (Self-assessment): hazardous for water
 Do not allow product to reach ground water, water course or sewage system.
 Must not reach sewage water or drainage ditch undiluted or unneutralised.
 Danger to drinking water if even small quantities leak into the ground.
 Harmful to aquatic organisms

· Results of PBT and vPvB assessment

This substance/mixture does not contain any components that can be considered persistent, bioaccumulative and toxic (PBT) or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

· PBT:

-

· vPvB:

-

· Other adverse effects

No further relevant information available.

*** 13 Disposal considerations****· Waste treatment methods****· Recommendation**

Must not be disposed together with household garbage. Do not allow product to reach sewage system.

This product, when disposed of in its unused and uncontaminated state, must be treated as hazardous waste according to EC Directive 2008/98/EC. Disposal practices must comply with all national and provincial laws and any municipal or local bylaws regarding hazardous waste. Additional evaluations may be required for used, contaminated and residual material. Do not discharge into sewers, soil or surface water. Incineration in a licensed incinerator suitable for this hazardous waste is recommended for this material. Small amounts of waste can be neutralized with e.g. polyol in preference to landfilling. Empty drums should be cleaned first (see Section 6) and then either punctured and scrapped or given to an approved reprocessor.

· Uncleaned packaging:**· Recommendation:**

Empty contaminated packagings thoroughly. They may be recycled after thorough and proper cleaning.

· Recommended cleansing agents:

Water, if necessary together with cleansing agents.

WE4

(Contd. on page 11)

Trade name: Herculan BA 2000 { B }

(Contd. of page 10)

* 14 Transport information

· UN-Number	
· ADR, IMDG, IATA	UN2735
· UN proper shipping name	
· ADR	2735 AMINES, LIQUID, CORROSIVE, N.O.S. (1,3-Benzenedimethanamine, ISOPHORONEDIAMINE)
· IMDG, IATA	AMINES, LIQUID, CORROSIVE, N.O.S. (1,3-Benzenedimethanamine, ISOPHORONEDIAMINE)
· Transport hazard class(es)	
· ADR	
	
· Class	8 (C7) Corrosive substances.
· Label	8
· IMDG, IATA	
	
· Class	8 Corrosive substances.
· Label	8
· Packing group	
· ADR, IMDG, IATA	II
· Environmental hazards:	Product contains environmentally hazardous substances: Fatty acids, C18 unsaturated, reaction products with triethylenetetramine
· Marine pollutant:	No
· Special precautions for user	Warning: Corrosive substances.
· Hazard identification number (Kemler code):	80
· EMS Number:	F-A,S-B
· Segregation groups	(SGG18) Alkalis
· Stowage Category	A
· Segregation Code	SG35 Stow "separated from" SGG1-acids
· Transport in bulk according to Annex II of Marpol and the IBC Code	Not applicable
· Transport/Additional information:	<----- aircraft ----->
· ADR	
· Excepted quantities (EQ):	E2
· Limited quantities (LQ)	1L
· Excepted quantities (EQ)	-
	Maximum net quantity per inner packaging: 30 g Maximum net quantity per outer packaging: 500 g
· Transport category	2
· Tunnel restriction code	E
· IMDG	
· Limited quantities (LQ)	1L

(Contd. on page 12)

WE4

Trade name: **Herculan BA 2000 { B }**

(Contd. of page 11)

· **Excepted quantities (EQ)**

Code: E2

Maximum net quantity per inner packaging: 30 g

Maximum net quantity per outer packaging: 500 g

· **UN "Model Regulation":**

UN 2735 AMINES, LIQUID, CORROSIVE, N.O.S. (1,3-BENZENEDIMETHANAMINE, ISOPHORONEDIAMINE), 8, II

* **15 Regulatory information**· **Safety, health and environmental regulations/legislation specific for the substance or mixture**· **Directive 2012/18/EU**· **Named dangerous substances - ANNEX I**

None of the ingredients is listed.

· **Seveso category**

Specified in regulation: Not applicable

· **National regulations:**· **Waterhazard class:**

Water hazard class 2 (Self-assessment): hazardous for water.

· **Other regulations, limitations and prohibitive regulations**

General international information on combating damage caused by fire and leakage.

www.ERICARDS.net

ERIC: 8-28

· **Chemical safety assessment:**

A Chemical Safety Assessment has not been carried out.

* **16 Other information**

This information is based on our present knowledge. However, this shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship.

· **Relevant phrases**

H302 Harmful if swallowed.

H312 Harmful in contact with skin.

H314 Causes severe skin burns and eye damage.

H315 Causes skin irritation.

H317 May cause an allergic skin reaction.

H318 Causes serious eye damage.

H319 Causes serious eye irritation.

H332 Harmful if inhaled.

H400 Very toxic to aquatic life.

H402 Harmful to aquatic life.

H410 Very toxic to aquatic life with long lasting effects.

H411 Toxic to aquatic life with long lasting effects.

H412 Harmful to aquatic life with long lasting effects.

· **Classification according to Regulation (EC) No 1272/2008**

The classification of the mixture is based on the calculation method using substance data according to Regulation (EC) No 1272/2008.

The safety data sheet is prepared in accordance with Regulation (EC) No. 1907/2006 (REACH) § 31, taking into account Annex II, which was amended by Commission Regulation (EU) 2020/878.

· **Department issuing SDS:**

Environmental Department

· **Contact:**

Dr. Michael Kissel

· **Abbreviations and acronyms:**

RID: Règlement international concernant le transport des marchandises dangereuses par chemin de fer (Regulations Concerning the International Transport of Dangerous Goods by Rail)

ICAO: International Civil Aviation Organisation

ADR: Accord relatif au transport international des marchandises dangereuses par route (European Agreement Concerning the International Carriage of Dangerous Goods by Road)

IMDG: International Maritime Code for Dangerous Goods

IATA: International Air Transport Association

EINECS: European Inventory of Existing Commercial Chemical Substances

ELINCS: European List of Notified Chemical Substances

CAS: Chemical Abstracts Service (division of the American Chemical Society)

DNEL: Derived No-Effect Level (REACH)

PNEC: Predicted No-Effect Concentration (REACH)

LC50: Lethal concentration, 50 percent

(Contd. on page 13)

Trade name: Herculan BA 2000 { B }

(Contd. of page 12)

LD50: Lethal dose, 50 percent
 PBT: Persistent, Bioaccumulative and Toxic
 SVHC: Substances of Very High Concern
 vPvB: very Persistent and very Bioaccumulative
 TSCA: Toxic Substances Control Act
 Acute Tox. 4: Acute toxicity – Category 4
 Acute Tox. 5: Acute toxicity – Category 5
 Skin Corr. 1B: Skin corrosion/irritation – Category 1B
 Skin Corr. 1C: Skin corrosion/irritation – Category 1C
 Skin Irrit. 2: Skin corrosion/irritation – Category 2
 Eye Dam. 1: Serious eye damage/eye irritation – Category 1
 Eye Irrit. 2: Serious eye damage/eye irritation – Category 2
 Skin Sens. 1: Skin sensitisation – Category 1
 Skin Sens. 1A: Skin sensitisation – Category 1A
 Skin Sens. 1B: Skin sensitisation – Category 1B
 Aquatic Acute 1: Hazardous to the aquatic environment - acute aquatic hazard – Category 1
 Aquatic Acute 3: Hazardous to the aquatic environment - acute aquatic hazard – Category 3
 Aquatic Chronic 1: Hazardous to the aquatic environment - long-term aquatic hazard – Category 1
 Aquatic Chronic 2: Hazardous to the aquatic environment - long-term aquatic hazard – Category 2
 Aquatic Chronic 3: Hazardous to the aquatic environment - long-term aquatic hazard – Category 3

· * Data compared to the previous version altered.